Summary in Graph

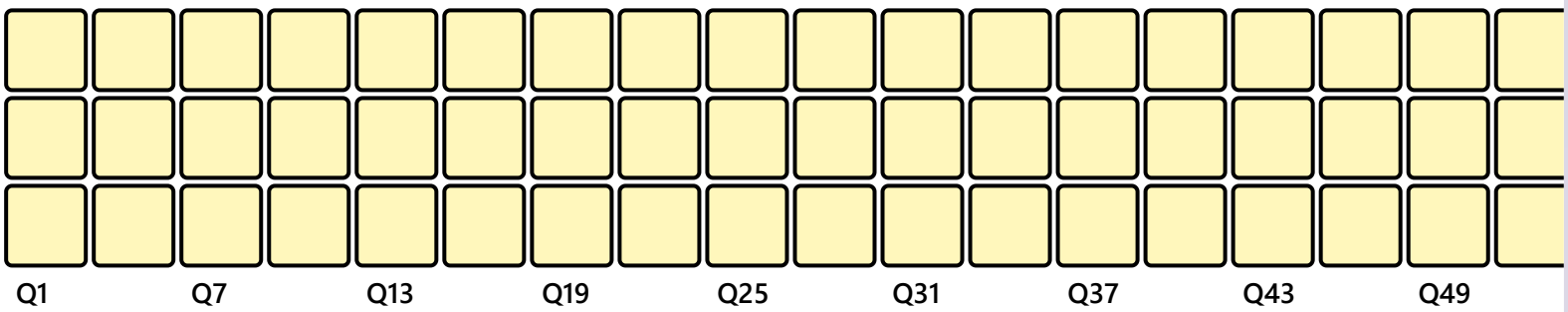
Exam Summary (GATE Overflow CS Test Series 2026 | Mock GATE 6)

Qs. Attempted:	0 (0 + 0)
Correct Attempts:	0 (0 + 0)
Incorrect Attempts:	0 (0 + 0)
Correct Marks:	0 (0 + 0)
Penalty Marks:	0 (0 + 0)
Resultant Marks:	0 (0 + 0)

Total Questions:	65 (30 + 35)
Total Marks:	100 (30 + 70)
Exam Duration:	180 Minutes
Time Taken:	0 Minutes

Individual Question Statistics

Select Metric: Marks Obtained



* Average Stats based on first-attempt stats of Top 10% (3) out of 34 users

EXAM RESPONSE EXAM STATS FEEDBACK

Aptitude

Q #1 Multiple Choice Type Award: 1 Penalty: 0.33 Verbal Aptitude

He has ___ a lot of money for his old age.

A. put off
B. put across
C. put aside
D. put through

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #2 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Verbal Aptitude

From among the given options choose the one closest in meaning to the given word

Impulsive

A. rash
B. deliberate
C. wary
D. impressive

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #3 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Verbal Aptitude

It is a standard practice _____ a civil servant to sign checks issued by the Treasury Department to ensure that government payments are not biased.

A. for
B. to
C. on
D. upon

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #4 | Numerical Type | Award: 1 | Penalty: 0 | Quantitative Aptitude

The probability that it is Monday and a student is absent is 0.08. Assuming there are 5 school days in a week, what is the probability that a student is absent given that today is Monday?

Your Answer: | Correct Answer: 0.4 | Not Attempted | Time taken: 00min 00sec | Discuss

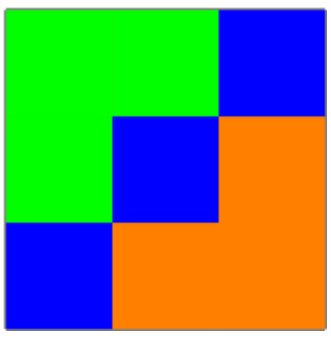
Q #5 | Numerical Type | Award: 1 | Penalty: 0 | Quantitative Aptitude

If the area of each of the 6 faces of a cuboid is grown by a factor of 16, the volume of the cuboid will grow by a factor of _____.

Your Answer: | Correct Answer: 64 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #6 | Numerical Type | Award: 2 | Penalty: 0 | Quantitative Aptitude

The number of different ways there are to color a 3×3 grid with green orange and blue paints, using each color 3 times is _____



Your Answer: | Correct Answer: 1680 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #7 Multiple Select Type | Award: 2 | Penalty: 0 | Quantitative Aptitude

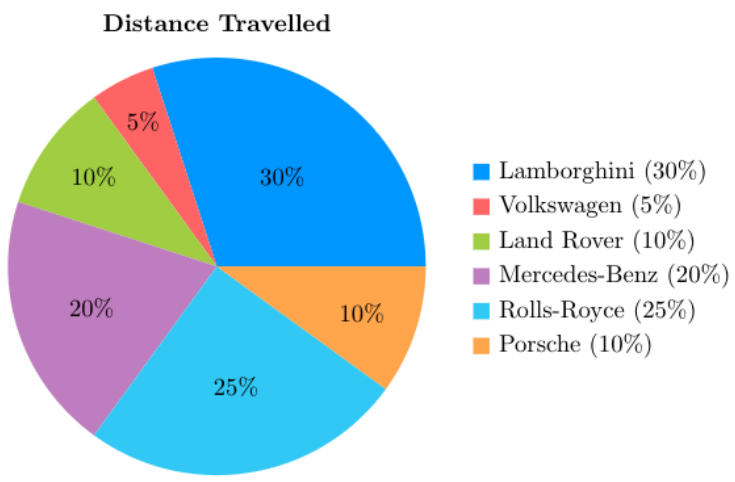
The sum of the series $1 \cdot 2 + 2 \cdot 2^2 + 3 \cdot 2^3 + \cdots + 2021 \cdot 2^{2021}$ is ____ (Mark all the appropriate choices)

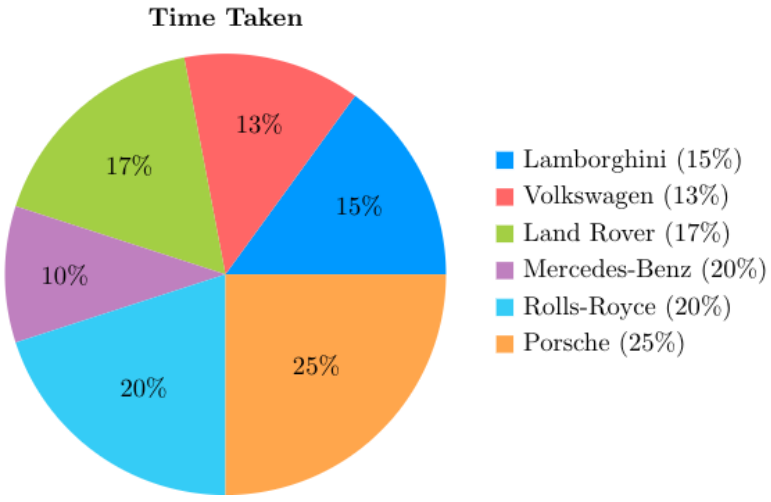
A. $4042 \cdot 2^{2020} + 2$
B. $4040 \cdot 2^{2021} + 2$
C. $2020 \cdot 2^{2022} + 2$
D. $2021 \cdot 2^{2019} + 2$

Your Answer: | Correct Answer: B;C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #8 Multiple Choice Type | Award: 2 | Penalty: 0.67 | Quantitative Aptitude

The following pie charts show the percentage of distance (in km) traveled by different cars and percentage of time (in hours) taken by different cars:





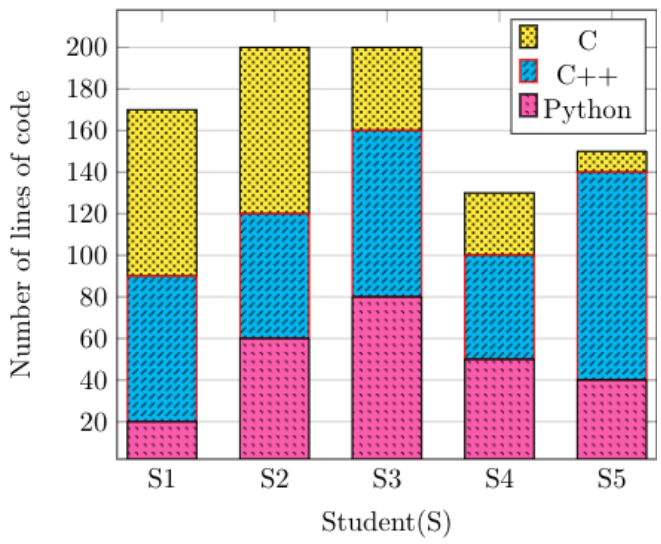
If the total time taken by all the cars is 60 hours and the difference between the speed of Lamborghini and Rolls-Royce is 10 km/hr, then the distance traveled by Mercedes-Benz?

A. 1080 km
B. 720 km
C. 160 km
D. 144 km

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 01sec | Discuss

Q #9 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Quantitative Aptitude

The bar graph below shows the programming output of five students over one month. The students write code in three different languages: C, C++, and Python.



Consider the following statements:

i. The number of lines of C++ code written by student $S3$ is less than the number of lines of C code written by $S5$ and the number of lines of Python code written by student $S4$.
ii. The total number of lines of Python code written by all students is greater than the total number of lines of C code written by them.

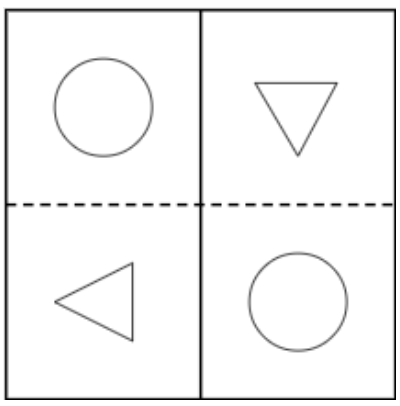
Which one of the above sentences is/are FALSE?

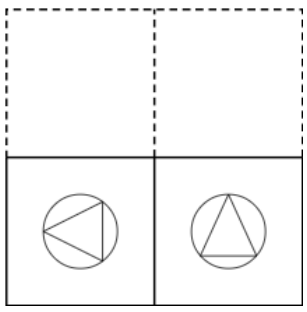
A. Only (i)
B. Only (ii)
C. Both (i) and (ii)
D. Neither (i) nor (ii)

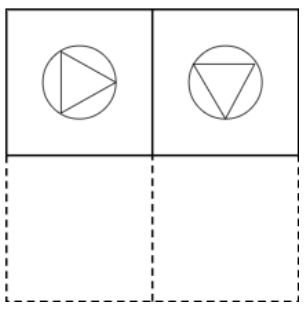
Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

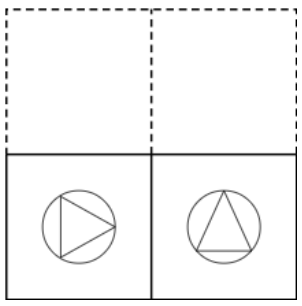
Q #10 | Multiple Select Type | Award: 2 | Penalty: 0 | Spatial Aptitude

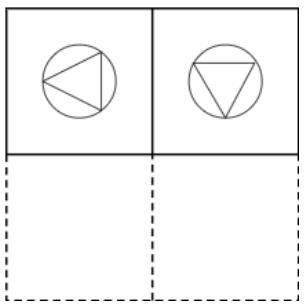
Find out amongst the four alternatives as to how the pattern would appear when transparent sheet is folded at the dotted line. (Mark all the appropriate choices)



A. 

B. 

C. 

D. 

Your Answer: | Correct Answer: A;D | Not Attempted | Time taken: 00min 00sec | Discuss

Technical

Q #1 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Graph Theory

Which of the following is TRUE regarding perfect matching in a tree?

A. A tree can have more than two perfect matching
B. A tree cannot have any perfect matching
C. A tree can have at most one perfect matching
D. A tree can have at most two perfect matching

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #2 | Numerical Type | Award: 1 | Penalty: 0 | Linear Algebra

The trace of the following matrix is _____

$$\begin{bmatrix} 3 & 3 & 3 & 3 & 3 \\ 3 & 2 & 2 & 2 & 3 \\ 3 & 2 & 1 & 2 & 3 \\ 3 & 2 & 2 & 2 & 3 \\ 3 & 3 & 3 & 3 & 3 \end{bmatrix}$$

Your Answer: | Correct Answer: 11 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #3 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Linear Algebra

What is the minimum value of $x^T Ax$ subject to $x^T x = 1$, if $A = \begin{bmatrix} 2022 & 2021 \\ 2021 & 2022 \end{bmatrix}$.

A. -1
B. 4043
C. 1
D. -4043

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #4 | Multiple Select Type | Award: 1 | Penalty: 0 | Mathematical Logic

Consider the following logical statements:

- P_1 : If budget happens or if gold prices go up, stock prices go down.
- P_2 : Stock prices went down.

Which of the following is NOT a valid conclusion? (Mark all the appropriate choices)

A. Gold prices went up
B. Gold prices went down
C. Budget happened
D. If budget did not happen then gold prices had gone down

Your Answer: | Correct Answer: A;B;C;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #5 | Numerical Type | Award: 1 | Penalty: 0 | Combinatory

A decimal number is called "increasing" if each digit is greater than the previous one (e.g., 1267 is one increasing number). The total number of 5 digit increasing numbers is _____

Your Answer: | Correct Answer: 126 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #6 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Graph Theory

A simple graph G has degree sequence $(7, 7, 6, 5, 5, 4, 3, 3, 2)$. What is the degree sequence of G^c?

A. $(2, 3, 3, 4, 5, 5, 6, 7, 7)$
B. $(1, 1, 3, 3, 4, 5, 5, 6, 6)$
C. $(1, 1, 2, 3, 3, 4, 5, 5, 6)$
D. $(2, 2, 3, 4, 4, 5, 6, 6, 7)$

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #7 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Programming in C

What will be the output of the following C program:

```c
#include<stdio.h>
int main()
{
    char* my_words[] = {"Ravi", "Raj", "Meena", "Dhanush", "Manu"};
    char *p;
    p = my_words[3];
    printf("%s, ", p+3);
    printf("%c", *p+3);
}
```

A. nush, nush
B. nush, n
C. Possible Runtime Error
D. nush, G

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #8 | Multiple Select Type | Award: 1 | Penalty: 0 | Data Structures

Given two sorted lists of M and N elements respectively, the best case time complexity to make a binary search tree out of the combined $M+N$ elements is (Mark all the appropriate choices)

A. $\Theta(M)$
B. $\Theta(M+N)$
C. $\Theta((M+N)\log(M+N))$
D. $O(M+N)$

Your Answer: | Correct Answer: B;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #9 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Data Structures

What is the best time complexity for finding the lowest common ancestor for two keys a and b $(a < b)$ in a height balanced binary search tree of n? You can assume that both a and b exist in the tree.

A. $\Theta(n)$
B. $\Theta(n \log n)$
C. $O(\log n)$
D. $\Omega(n)$

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #10 | Multiple Select Type | Award: 1 | Penalty: 0 | Theory of Computation

A language L has the prefix-free property if there are no two distinct strings in L, such that one is a prefix of the other. Which of the following language classes have prefix-free property? (Mark all the appropriate choices)

A. Finite Languages
B. Language accepted by a Deterministic Pushdown Automata by Empty Stack
C. Language accepted by a Deterministic Pushdown Automata by Final State
D. The language generated by $\mathrm{LR}(0)$ grammar

Your Answer: | Correct Answer: B;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #11 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Theory of Computation

Consider the following language definition:
$L = \{\langle M \rangle \mid M$ is a Deterministic Finite Automation and $L(M)$ contains a non-recursively enumerable subset$\}$
L is

A. Recursively enumerable but not recursive
B. Recursive
C. Regular
D. Context-free

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #12 | Multiple Select Type | Award: 1 | Penalty: 0 | Databases

From the following instance of a relation schema $R(A, B, C)$, we can conclude that:(Mark all the appropriate options)

A	B	C
1	7	1
1	7	4
2	3	5
4	2	2
2	3	5

A. A functionally determines B and B functionally determines A.
B. A does not functionally determine C and B does not functionally determine C.
C. C functionally determines B.
D. C functionally determines A.

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #13 | Numerical Type | Award: 1 | Penalty: 0 | Computer Networks

Consider that 30 machines need to be connected in a LAN using 4-port Ethernet switches. Assume that these switches do not have any separate uplink ports. The minimum number of switches needed is _____

Your Answer: | Correct Answer: 15 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #14 | Multiple Select Type | Award: 1 | Penalty: 0 | Computer Networks

Which of the following is/are CORRECT? (Mark all the appropriate choices)

A. A repeater operates at the physical layer

B. A hub operates at the data-link layer
C. A switch operates at the data-link layer
D. A bridge operates at the network layer

Your Answer: | Correct Answer: A;C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #15 Numerical Type | Award: 1 | Penalty: 0 | Compiler Design

The least number of temporary variables required to create a three-address code in static single assignment form for the expression $a * b + c/d - 3$ is _____

Your Answer: | Correct Answer: 4 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #16 Multiple Select Type | Award: 1 | Penalty: 0 | Compiler Design

Which of the following statements regarding LR parsers is/are FALSE? (Mark all the appropriate choices)

A. A $\mathrm{LR}(0)$ configuration set cannot have multiple reduced items
B. A $\mathrm{LR}(0)$ configuration set cannot have both shifts as well as reduced items
C. A $\mathrm{SLR}(1)$ parser does not allow both SHIFT and REDUCE items to be in the same state
D. An $\mathrm{LR}(0)$ parser can parse any regular grammar

Your Answer: | Correct Answer: C;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #17 Multiple Select Type | Award: 1 | Penalty: 0 | Linear Algebra

If $A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 1 & 0 & 0 \end{bmatrix}$, then $A^{2027} - A^{2022} =$

A. $A^5 - A$
B. A^6
C. $A^7 - A^2$
D. $A^6 - A$

Your Answer: | Correct Answer: C;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #18 Multiple Select Type | Award: 1 | Penalty: 0 | Algorithms

Consider the following SQL query:

```
select studentid,name,age from STUDENT order by age,name;
```

Here, we need to sort the student data first by age and then by name. For the sorting by name which of the following algorithms is/are inappropriate to be used? (Mark all the appropriate choices)

A. Merge Sort
B. Insertion Sort
C. Quick Sort
D. Heap Sort

Your Answer: | Correct Answer: C;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #19 Numerical Type | Award: 1 | Penalty: 0 | Probability

A fair die is thrown as long as necessary for 6 to turn up. If the number of throws (n) turns out to be odd, the probability that $n = 5$ is _____ (rounded to 2 decimal points)

Your Answer: | Correct Answer: 0.14:0.15 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #20 Numerical Type | Award: 1 | Penalty: 0 | Probability

A stick of length 10 m is broken into two pieces, at a randomly chosen break point. The expected length of the shorter piece (in meters) is _____

Your Answer: | Correct Answer: 2.5 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #21 Multiple Choice Type | Award: 1 | Penalty: 0.33 | Programming in C

What will be the output of the following C program?

```c
#include<stdio.h>
#include<stdlib.h>
struct node
{
    int value;
    struct node * next;
};
struct node* init_list()
{
    struct node* list = malloc(sizeof(struct node));
    list -> value = 5;
    list -> value++;
    return list;
}
int main()
{
    struct node *list;
    list = init_list();
    printf("%d", list-> value);
}
```

A. 5
B. Garbage Value
C. 6
D. Possible Runtime Error

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #22 Multiple Select Type | Award: 1 | Penalty: 0 | Digital Logic

Which of the following functions is/are associative? (Mark all the appropriate choices)

A. AND
B. NOR
C. EXOR

D. NAND

Your Answer: | Correct Answer: A;C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #23 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Algorithms

Given an array of N distinct elements, you have to design an algorithm that should return K elements in sorted order. The returned K elements can be from any position in the given array. The least time complexity of such an algorithm assuming comparison based sorting will be

A. $\Theta(K \log N)$
B. $\Omega(\log N)$
C. $\Omega(K^2)$
D. $\Omega(K \log K)$

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #24 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Calculus

The value of $\int_0^{\frac{\pi}{2}} \sin^4 x \cos^4 x dx$ is ____

A. $\left(\frac{3\pi}{256}\right)$
B. $\left(\frac{5\pi}{768}\right)$
C. $\left(\frac{7\pi}{768}\right)$
D. $\left(\frac{3\pi}{384}\right)$

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #25 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Probability

An unbiased coin is tossed repeatedly and its outcome is recorded. What is the expected number of trials to get a Head followed by a Tail?

A. 6
B. 5
C. 4
D. 3

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #26 | Multiple Select Type | Award: 2 | Penalty: 0 | Graph Theory

Which of the following statement is/are correct? (Mark all the appropriate choices)

A. In every k-chromatic graph, there are at least k vertices of degree at least $k-1$.
B. A Hamiltonian graph can have pendant edge.
C. A graph with number of vertices less than 5 can be non-planar.

D. A graph with number of edges less than 9 can be non-planar.

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #27 Numerical Type | Award: 2 | Penalty: 0 | Set Theory & Algebra

Consider set $A = \{1, 2, 3, 4\}$. From all the subsets of A, we define a new set X as $X = \{(P, Q) \mid |P \Delta Q| = 2\}$, where (P, Q) is ordered pair (P, Q) and Δ denotes the symmetric difference operation. The number of elements in X, i.e., $|X|$ is _____

Your Answer: | Correct Answer: 96 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #28 Numerical Type | Award: 2 | Penalty: 0 | Combinatory

Billionaire Ram has 5 distinct properties which he decided to distribute among his 3 daughters. He hires a consultancy firm for the same which charges Rs. 500 for each of the possible distribution assuming each daughter gets at least one property and all daughters being considered distinct. How much fee in rupees will Ram have to pay the agency?

Your Answer: | Correct Answer: 75000 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #29 Multiple Select Type | Award: 2 | Penalty: 0 | Digital Logic

Which of the following decimal number does not cause a loss of precision when represented in IEEE-754 single-precision representation? (Mark all the appropriate choices)

A. $4096 \times 4096 \times 4096$
B. $10 \times 10 \times 10$
C. $256 \times 256 \times 256 \times 256$
D. $8 \times 8 \times 8$

Your Answer: | Correct Answer: A;B;C;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #30 Multiple Select Type | Award: 2 | Penalty: 0 | Databases

Which of the following statements concerning database normalization is/are TRUE? (Mark all the appropriate choices)

A. A necessary condition for a relation to be in $1\ \mathrm{NF}$ but not in $2\ \mathrm{NF}$ is that at least one of its candidate keys must have multiple attributes.
B. A necessary condition for a relation to be in $3\ \mathrm{NF}$ but not in BCNF is that some prime attribute must be determined by a non-key.
C. A necessary condition for a relation to be in $2\ \mathrm{NF}$ but not in $3\ \mathrm{NF}$ is that all non-prime attributes must be determined by a non-key.
D. A necessary condition for a relation to be in $3\ \mathrm{NF}$ but not in BCNF is that it should have overlapping candidate keys.

Your Answer: | Correct Answer: A;B;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #31 Numerical Type | Award: 2 | Penalty: 0 | Digital Logic

The literal count of a Boolean expression is the sum of the number of times each literal appears in the expression. For example, the literal count of $(\overline{x}y + x\overline{z} + \overline{y}z)$ is 6. If the minimum possible literal counts of the sum-of-product and product-of-sum representations of the function $f(x_1, x_2, x_3, x_4) = \Sigma_m(0, 1, 2, 3, 7, 8, 10) + \Sigma_d(4, 5, 6, 11, 15)$ are given by α and β respectively, the value of $\alpha + 2\beta =$_____

Your Answer: | Correct Answer: 11 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #32 Numerical Type | Award: 2 | Penalty: 0 | CO & Architecture

Consider a 5-stage pipeline - IF (Instruction Fetch), ID (Instruction Decode and register read), EX (Execute), MEM (Memory), and WB (Write Back). All register reads take place in the second phase of a clock cycle and all register writes occur in the first phase and there is no operand forwarding in use. Consider the execution of the following instruction sequence:

- $I_1 : R_1 \leftarrow R_2 + R_3$
- $I_2 : R_3 \leftarrow R_1 - R_2$
- $I_3 : M[R_3] \leftarrow R_1$
- $I_4 : R_2 \leftarrow R_3 * R_1$

If the number of RAW (Read after write) hazards is denoted by A, WAR (Write after read) hazards by B and WAW (Write after write) hazards by C, then $2A + 3B + C =$ _____

Your Answer: | Correct Answer: 8 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #33 Numerical Type | Award: 2 | Penalty: 0 | CO & Architecture

A Hard disk is connected to a 2 GHz processor through a DMA controller. Assume that the initial setup time for a DMA transfer takes 800 clock cycles for the processor and also assume that the handling of the interrupt on DMA completion requires 400 clock cycles for the processor. The hard disk has a transfer rate of 4000 Mbps and the average block size transferred is 4 Kilobytes. The fraction of the processor time consumed by the disk, assuming that the data are transferred only during the idle cycles of the CPU is _____

Your Answer: | Correct Answer: 0.06:0.07 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #34 Numerical Type | Award: 2 | Penalty: 0 | CO & Architecture

Consider a CPU with an average CPI of 1.6 when all memory accesses hit on the cache. Assume an instruction mix

ALU	55%
LOAD	30%
STORE	10%
BRANCH	5%

If the cache miss rate is 3.5% and miss penalty is 60 cycles, calculate the effective CPI if $L1$ cache uses a write back and write allocate policy with the probability of a cache block being dirty being 0.1 (round off to 2 decimal places). Assume no penalty for instruction fetches working via a separate instruction cache.

Your Answer: | Correct Answer: 2.52:2.53 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #35 Numerical Type | Award: 2 | Penalty: 0 | CO & Architecture

Consider a 5 stage MIPS Pipeline with the following features:

- For a non-pipeline implementation, ALU and branch instructions take 3 cycles while memory operations take 5 cycles and the clock rate is $2\ \mathrm{GHz}$.
- Relative frequencies of ALU, branch, and memory instructions are $65\%, 15\%$, and 20%.
- In the pipeline due to clock skew and setup time, the clock cycle time increases by $0.15\ \mathrm{ns}$.

Calculate the estimated speedup due to pipelining assuming 0.1% branch misprediction rate and 100 cycles branch misprediction penalty. (rounded to 2 decimal points).

Your Answer: | Correct Answer: 2.3:2.4 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #36 Numerical Type | Award: 2 | Penalty: 0 | CO & Architecture

A device with a transfer rate of 50 kilobytes per second is connected to a CPU. Data is transferred byte-wise and the interrupt overhead is 5 microseconds. The byte transfer time between the device interface register and CPU or memory is negligible. What is the minimum performance gain of an operating device under interrupt-driven mode over operating it under programmed IO mode?

Your Answer: | Correct Answer: 4 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #37 Numerical Type | Award: 2 | Penalty: 0 | Data Structures

In a binary search tree, the following key values (not necessarily in the order given) are encountered while searching for the key 24.

$$1, 5, 9, 13, 17, 21, 25, 29, 33.$$

The total number of possible orders in which the given keys of the binary search tree could have been visited is _____

Your Answer: | Correct Answer: 84 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #38 Numerical Type | Award: 2 | Penalty: 0 | Algorithms

You need to store 4 records in a hash table of size 10 using open addressing with linear probing. The hash function in use has the probability that a key is hashed into any of the 10 slots as $1/10$. For a particular slot in the hash table, what is the probability that this slot is empty, that is, none of the 4 keys hashes into this slot (rounded to 1 decimal point)?

Your Answer: | Correct Answer: 0.6 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #39 Multiple Select Type | Award: 2 | Penalty: 0 | Algorithms

Which of the following statements is/are TRUE?(Mark all the correct options)

A. The tightest upper bound on the worst case time complexity in determining the existence of a cycle in an undirected graph $G = (V, E)$ is $\Theta(|E|)$.
B. If in a depth-first traversal of a graph G with n vertices, k edges are marked as tree edges, then the number of connected components in G is $n - k$.

C. The tightest upper bound on the worst case time complexity in determining the existence of a cycle in an undirected graph $G = (V, E)$ is $O(|V|)$.
D. In a connected graph $G = (V, E)$ with n vertices DFS will yield $n - 1$ tree edges.

Your Answer: | Correct Answer: B;C;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #40 Multiple Select Type | Award: 2 | Penalty: 0 | Algorithms

A cut $S = (V - S)$ of an undirected graph $G = (V, E)$ is a partition of V. We say that an edge $(u, v) \in E$ crosses the cut $(S, V - S)$ if one of its endpoints is in S and the other is in $V - S$. An edge is a light edge crossing a cut if its weight is the minimum of any edge crossing the cut. Considering undirected graphs with all distinct edge weights mark all the correct options given below.

A. $\{(u, v)$: there exists a cut$(S, V - S)$ such that (u, v) is a light edge crossing it $\}$ forms a MST.
B. If an edge is contained in some Minimum Spanning Tree (MST), it is a light edge crossing some cut.
C. A graph G has a unique MST if, for every cut, there is a unique light edge crossing the cut.
D. If a graph has a unique MST, then for every cut of the graph there is a unique light edge crossing the cut.

Your Answer: | Correct Answer: A;B;C;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #41 Multiple Select Type | Award: 2 | Penalty: 0 | Theory of Computation

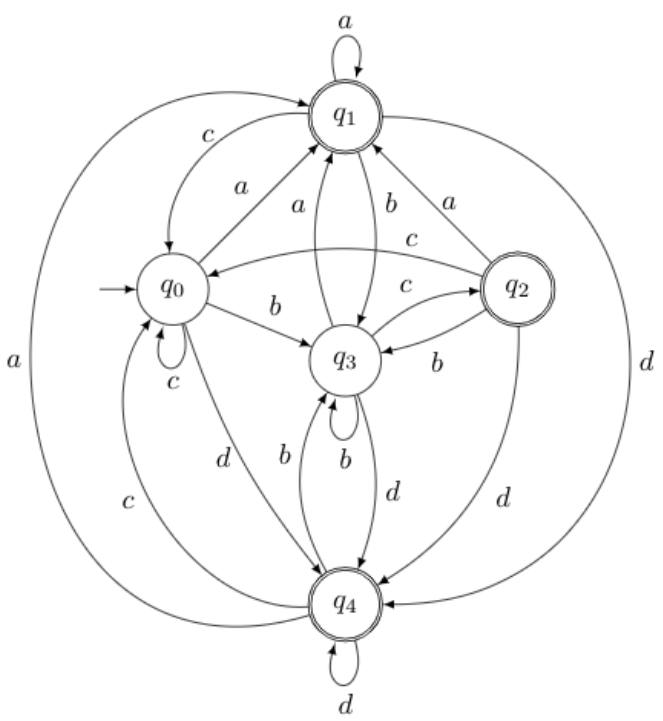
Which of the following languages is/are NOT regular over $\Sigma = \{a, b, c\}$? (Mark all the appropriate choices)

A. $L = \{xcww_R \mid w, x \in (a + b)^*\}$
B. $L = \{xcww \mid w, x \in (a + b)^*\}$
C. $L = \{xcwyw \mid w, x, y \in (a + b)^*\}$
D. $L = \{wxcw_R \mid w, x \in (a + b)^*\}$

Your Answer: | Correct Answer: A;B;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #42 Multiple Select Type | Award: 2 | Penalty: 0 | Theory of Computation

Which of the given regular expressions generate the same language as the following Deterministic Finite Automaton over $\Sigma = \{a, b, c, d\}$? (Mark all the appropriate choices)



A. $(a^+ + b^+c + c^*d^+)^+$

B. $(a^* + b^+c + c^*d^+)^+$
C. $(aa^* + b^+c + c^*d^+)^+$
D. $(a^+ + b^+c + c^*d^*)^+$

Your Answer: | Correct Answer: A;C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #43 Multiple Choice Type | Award: 2 | Penalty: 0.67 | Databases

The student information in a college is stored in the following relation:
student(Name, Sex, Course, CGPA, Department)

Consider the following SQL query and choose the correct option:

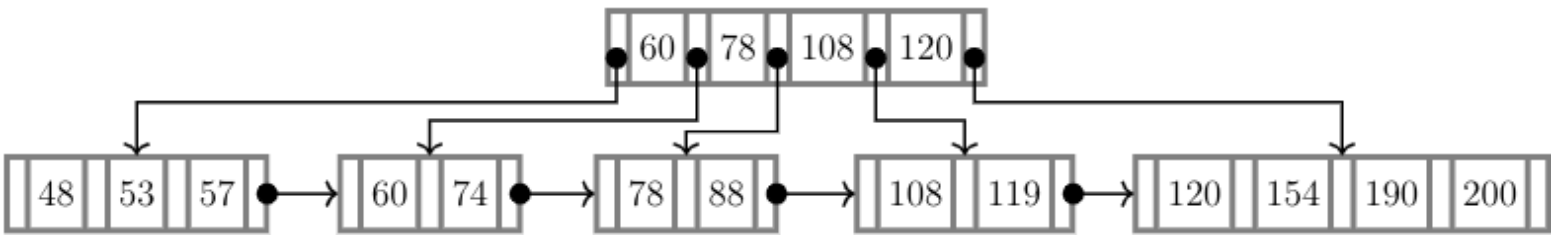
```
   SELECT Department FROM student
   WHERE sex='F'
   GROUP BY Department
   HAVING AVG(CGPA)>(SELECT AVG(CGPA)
5. FROM student GROUP BY Department
   HAVING COUNT(*) > 50);
```

A. The query returns the Departments in which the average CGPA of all students is greater than the average CGPA of female students from all departments having more than 50 students.
B. The query returns the Departments in which the average CGPA of female students is greater than the average CGPA of students from all departments having more than 50 students.
C. The query returns the Departments in which the average CGPA of female students is greater than the average CGPA of students from all departments having more than 50 female students.
D. The query returns the Departments in which the average CGPA of female students is not less than the average CGPA of students from all departments having more than 50 female students.

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #44 Multiple Choice Type | Award: 2 | Penalty: 0.67 | Databases

The B^+ tree of order 5 and having 6 nodes is shown in the following figure.



A value of 125 is inserted into the B^+ tree, and then the key 200 is deleted from the B^+ tree. After these two operations, the number of nodes in the final B^+ tree will be

A. 5
B. 6
C. 7
D. 9

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #45 Numerical Type | Award: 2 | Penalty: 0 | Computer Networks

In a packet switched network, packets are routed from source to destination along a single path having 2 intermediate nodes. If the message size is 32 bytes and each packet contains a header of 4 bytes, then the

optimum packet size including header in bytes is _____

Your Answer: | Correct Answer: 12 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #46 | Numerical Type | Award: 2 | Penalty: 0 | Computer Networks

Consider the sliding window flow-control protocol operating between a sender and a receiver over a full-duplex error-free link. Assume the following:

- The size of the data frame is $4,000$ bits and the size of the acknowledgment frame is 20 bits.
- The link data rate in each direction is 2 Mbps ($= 2 \times 10^6$ bits per second).
- The one-way propagation delay of the link is 100 milliseconds.

If the frame processing times are negligible and the sender has an infinite number of frames available for transmission, the minimum value of the sender's window size in terms of the number of frames, (rounded to the nearest integer) is needed to achieve a link utilization of 80% is _____

Your Answer: | Correct Answer: 81 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #47 | Numerical Type | Award: 2 | Penalty: 0 | Computer Networks

IIT Bombay and IISc Bangalore are connected by a 1 Giga Ethernet link with a round trip latency of 50 milliseconds. If an IISc student tries to transfer a large file from a server at IISc. to a server in IIT Bombay using FTP, what is the best throughput in bits per second (rounded to 2 decimal points) she can expect assuming standard 64 KB TCP window size?

Your Answer: | Correct Answer: 10485760 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #48 | Numerical Type | Award: 2 | Penalty: 0 | Operating System

Consider the following code segment:

```
pid_t pid;
pid = fork();
if (pid != 0) {
    thread create(...);
    fork();
}
else {
    fork();
}
```

If a is the total number of processes created, and b is the total number of threads created when the above code is executed, the value of $2a + 3b =$ _____

Your Answer: | Correct Answer: 23 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #49 | Multiple Select Type | Award: 2 | Penalty: 0 | Operating System

Consider the following synchronization solution.

```
    do {
        key = TRUE;
        while ( key == TRUE) atomic_swap (&lock, &key);
        <Critical Section>
5.      lock = FALSE;
    } while (TRUE);
```

Here, atomic_swap is an atomic version of swap. If the boolean shared variable lock is initialized to FALSE, all other variables are local to the processes and all the assignment statements happen atomically in hardware, which of the following statements is/are TRUE? (Mark all the appropriate choices)

A. Race condition may be possible
B. Mutual exclusion is guaranteed
C. Progress is guaranteed
D. Starvation freedom is guaranteed

Your Answer: | Correct Answer: B;C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #50

Numerical Type | Award: 2 | Penalty: 0 | Operating System

A demand paging system takes 500 time units to service a page fault when page is not dirty and 501 time units in case the page is dirty. Memory access time is 1 time unit. The probability of a page fault is p. In case of a page fault, the probability of the page being dirty is also p. It is observed that the average access time is 12 time units. Then the value of $100p$ is _____ (up to two decimal places).

Your Answer: | Correct Answer: 2.1:2.3 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #51

Numerical Type | Award: 2 | Penalty: 0 | Operating System

A typical program has 30% memory instructions. Assume there are 4% data TLB misses, each requiring 100 extra cycles to be handled. Also assume that each instruction requires 1 cycle to execute, TLB access takes 1 cycle, each memory operation in the physically addressed cache takes 1 cycle, 25% of data accesses are cache misses, and each cache miss takes 75 extra cycles to be handled. How long (in cycles) would it take to execute 1000 instructions?

Your Answer: | Correct Answer: 8425 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #52

Multiple Choice Type | Award: 2 | Penalty: 0.67 | Operating System

Consider a CPU using 32 bit address space. The size of the physical memory in use is 256 MB and the page size is 4 Kilo bytes. If the size of a Page Table Entry is four bytes for standard page table and eight bytes for inverted page table, the sizes of the page tables (assuming single level) required for both standard and inverted page tables respectively will be

A. 2^{20} bytes, 2^{15} bytes
B. 2^{20} bytes, 2^{15} bytes
C. 2^{18} bytes, 2^{22} bytes
D. 2^{22} bytes, 2^{19} bytes

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #53

Numerical Type | Award: 2 | Penalty: 0 | Compiler Design

Consider the following program.

```
int x = 10;
int main() {
    x = 1;
    f();
    g();
}
void f() {
    int x = 3;
    h();
}
void g() {
    x = 2;
    h();
}
void h() {
    printf("%d\n",x);
}
```

If α and β are the sum of all the values printed by the above code assuming static and dynamic scoping respectively, then $2\alpha + 3\beta =$ _____

Your Answer: | Correct Answer: 21 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #54

Multiple Choice Type | Award: 2 | Penalty: 0.67 | Programming in C

What will be the output of the following C code?

```
#include <stdio.h>
int main() {
    int sum = 0 ;
    for(int i = 0; i < 16; i++) {
        char *p = (char*)&i;
        for(int j = 0; j < sizeof(int); j++) {
            sum += *p++;
        }
    }
    printf("%d", sum);
}
```

A. 120
B. 16
C. 9
D. Garbage Value

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #55

Multiple Select Type | Award: 2 | Penalty: 0 | Compiler Design

Consider the following grammar:

- $S \rightarrow aT$
- $T \rightarrow aT \mid \epsilon$

The above grammar is _____ (Mark all the appropriate choices)

A. $\mathrm{LL}(1)$
B. $\mathrm{SLR}(1)$
C. $\mathrm{LALR}(1)$
D. $\mathrm{LR}(1)$

Your Answer: | Correct Answer: A;B;C;D | Not Attempted | Time taken: 00min 00sec | Discuss

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